

Xuanyou Chen

Tel: (760) 390-0895 | Email: xchen3019@gatech.edu | [Personal Website](#) | F-1 Student

Education - Dual Degree Program

Georgia Institute of Technology | Atlanta, GA

Aug 2025 - May 2027

Bachelor of Science in Computer Engineering

GPA: **4.0/4.0**

Specializations: Distributed Systems & Software Design; Robotics & Autonomous Systems

Relevant Coursework: Advanced Robotic Manipulation, Introduction to Automation & Robotics, Introduction to Perception & Robotics, Feedback Control Systems, Machine Learning, Introduction to Artificial Intelligence

Emory University | Atlanta, GA

Aug 2022 - May 2025

Bachelor of Science in Computer Science and Mathematics

GPA: **4.0/4.0**

Skills

Programming Languages: Python, C++, C, MATLAB, Java, JavaScript/TypeScript, SQL

Robotics & Motion Planning: ROS 2, MuJoCo, Drake, OMPL, MoveIt, LeRobot, URDF

Machine Learning, Perception & 3D Geometry: PyTorch, OpenCV, Open3D, NumPy, SciPy, VGGT, SAM2, GroundingDINO, FoundationPose, AnyGrasp, Trimesh, Hunyuan3D

Software Engineering: Linux, Git, React, MySQL, MongoDB, AWS, Google Cloud

Hardware & Prototyping: KUKA LBR iiwa 7 R800, AgileX PiPER, LeRobot SO-101, MyCobot-280, SCHUNK EGK-40, Intel RealSense D415, Orbbec-336L, camera calibration, 3D printing, circuit soldering

Research

Physically Verified Image-Editing-Based Trajectory Generation (*1st Author, submitted to ICRA 2027*)

Jan 2026 - Sep 2026

Tools: KUKA iiwa 7, ROS2, Drake, MuJoCo, OMPL

- Built a VLM-based task planner converting human instructions and RGB-D images into physically verified 6D object goal pose.
- Developed perception module using VGGT for depth estimation, GroundingDINO+SAM2+FoundationPose for pose estimation.
- Designed a novel VLM reasoning interface that identifies incorrect translational and rotational components, refines candidate poses, and physically validates them in MuJoCo.
- Utilized AnyGrasp and OMPL RRT-Connect to generate grasps and collision-free trajectories, executed through ROS2+Drake.
- Achieved 74.07% task-planning and 71.76% real-robot execution success across nine manipulation tasks (160/216 and 155/216 trials), outperforming OmniManip (43.98%) and ReKep (32.87%) in task planning; parallel-world validation improved mesh-free execution from 42.59% to 60.19% (92/216 to 130/216).

End-to-End Imitation Learning on LeRobot SO-101 | [Project Webpage](#)

Nov 2025 - June 2026

Tools: LeRobot SO-101, PyTorch, ACT, ResNet-18

- Built an end-to-end visuomotor IL system mapping RGB images and 6-D joint proprioception to 6-D joint/gripper commands.
- Trained policies across three datasets: 10 demos/5,557 samples, 50 demos/9,600 samples, and 100 demos/70,904 samples.
- Implemented an ACT-style policy with a pretrained ResNet-18 encoder and 4-layer Transformer to predict 30-step action chunks; deployed with receding-horizon inference every three control steps.
- Achieved bottle pick-and-place and regrasp recovery after addressing insufficient data, shadow-induced spurious correlations, poor camera viewpoint, and lack of proprioceptive feedback.

Experience

Intelligent Vision & Automation Laboratory (IVALab) | Robotics Infrastructure Developer

Aug 2025 - May 2026

- Built a Python package for **MyCobot-280** for kinematics, gripper control, trajectory recording, and camera calibration.
- Collaborated on ArUco-based camera calibration and implemented “click-to-capture” vision-guided picking pipeline.
- Customized the full hardware stack for a suction gripper, including a MOSFET-based pump/valve switching circuit, GPIO interfaces, and 3D-printed mechanical adapter for mounting onto the arm.
- Diagnosed and repaired two malfunctioning electric grippers—opened and serviced the hardware, coordinated with vendor technical support, and successfully secured two replacement servos at no cost.

RoboNav, RoboJackets | Software Team Member

Sep 2025 - Dec 2025

- Implemented in **ROS2** and **C++** camera-based obstacle detection using HSV filtering to generate top-down occupancy grids.
- Built key components of a SLAM pipeline: implemented odometry sensor model and particle-filter localization, and a mapping node that uses TF transforms to fuse obstacle detections into a global occupancy grid.
- Developed a hill-climbing optimization module that samples neighboring terrain and steers toward max-elevation goal points.

Emory Center for AI Learning | Project Leader

Jan 2025 - May 2025

- Led the development of a medical-device support chatbot for MedView, utilizing AI to answer device-related questions.
- Developed a React & TypeScript frontend supporting both text and voice interaction using the Web Speech API.
- Built a FastAPI backend to query the DeepSeek API, with semantic caching using Sentence Transformers and FAISS for low-latency FAQ retrieval from a predefined MongoDB database.

Applied AI Ventures Lab | Research Assistant

Jan 2024 - May 2024

- Developed a full-stack video analytics platform with Next.js frontend and a Spring Boot backend (42+ REST APIs).
- Designed a database schema in MySQL and deployed backend server on Linux-based AWS EC2 instances.
- Implemented a video content similarity model using GPT-4 & spaCy for transcript-based semantic analysis.

Curastone | Software Developer Intern

Sep 2023 - Dec 2023

- Assisted with the development of a Next.js AI learning assistant generating flashcards and personalized problem sets.
- Implemented user authentication, file upload, and course management, integrated with backend API using Redux.
- Designed responsive webpages using Tailwind CSS to ensure proper display of elements on various screen sizes.
- Deployed website using Vercel and AWS Route 53 and documented the deployment process for future reference.

Emory Robotics Club | Programming Team Member

Sep 2024 - Dec 2024

- Built and programmed the VEX V5 Classroom Starter Kit robot as a foundation for autonomous and driver control;
- Developed motion code using the PROS C++ library for the 2025 VEX “High Stakes” competition;
- Implemented path-planning logic and tuned PID controllers for drivetrain and lift stability.